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which exchanges $3Na^+$outwards for $2K^+$ into the cells. In this state, the cell membrane is said to be **polarized**. In neuron, the **resting membrane potential** ranges from -**40mV** to -**90mV**, and its normal value is -**70mV**. The minus sign indicates that the inside of the cell is negative with respect to the outside (Figure 10.4).

Action membrane potential

An action potential occurs when a neuron sends information down an axon, away from the cell body. It includes following phases, depolarization, repolarisation and hypo polarization.

Depolarization – Reversal of polarity

When a nerve fibre is stimulated, **sodium voltage-gate opens** and makes the **axolemma** permeable to Na^+ ions; meanwhile **the potassium voltage gate** closes. As a result, the rate of flow of Na^+ ions into the **axoplasm** exceeds the rate of flow of K^+ ions to the outside fluid [ECF]. Therefore, the axolemma becomes positively charged inside and negatively charged outside. This reversal of electrical charge is called **Depolarization**.

During depolarization, when enough Na^+ ions enter the cell, the action potential reaches a certain level, called **threshold potential** [-55mV]. The particular stimulus which is able to bring the membrane potential to threshold is called **threshold stimulus**.

The action potential occurs in response to a **threshold stimulus** but does not occur at **subthreshold stimuli**. This is called **all or none principle.** Due to the rapid influx of Na^+ ions, the membrane potential shoots rapidly up to +45mV which is called the **Spike potential**.

Repolarisation [Falling Phase]

When the membrane reaches the spike potential, **the sodium voltage-gate closes** and **potassium voltage-gate opens**. It checks influx of Na^+ions and initiates the efflux of K^+ions which lowers the number of positive ions within the cell. Thus, the potential falls back towards the resting potential. The reversal of membrane potential inside the axolemma to negative occurs due to the efflux of K^+ ions. This is called **Repolarisation**.

Hyperpolarization

If repolarization becomes more negative than the resting potential -70 mV to about -90 mV, it is called **Hyperpolarization**. During this, K^+ ion gates are more permeable to K^+ even after reaching the threshold level as it closes slowly; hence called **Lazy gates**. The membrane potential returns to its original **resting state** when K^+ ion channels close completely. During hyperpolarization the Na^+ voltage gate remains closed (Figure 10.5).

Conduction Speed of a nerve impulse

The conduction speed of a nerve impulse depends on the diameter of axon. The greater the axon's diameter, the faster is the conduction. The **myelinated axon** conducts the impulse faster than the **non-myelinated axon**. The voltage-gated Na^+ and K^+ channels are concentrated at the **nodes of Ranvier**. As a result, the impulse jumps node to node, rather than travelling the entire length of the nerve fibre. This mechanism of conduction is called **Saltatory Conduction**. Nerve impulses travel at the speed of 1-300 m/s.

10.3.3 Synaptic transmission

The junction between two neurons is called a **Synapse** through which a nerve